Rohit Kumar

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SUMMARY

Cloud and platform engineer with 9+ years of experience designing secure, scalable cloud infrastructure, control planes, internal developer platforms, and AI/ML-assisted operations tooling in large-scale enterprise environments. Builds reusable infrastructure as code with **Terraform** and **Python** and ships ideas from architecture to production through **CI/CD** automation. Proven track record of turning operational bottlenecks into durable engineering systems, including compressing cloud region launch timelines from **30 days to 8 hours**, lowering state-tier infrastructure usage by **68.4%**, and removing recurring on-call toil through automation and self-service. Deep expertise across distributed systems, infrastructure automation, reliability engineering, API design, serverless and event-driven backends, and data & AI platforms, with strong ownership of high-stakes production initiatives.

SKILLS

Languages : Java, Python, Shell Scripting
Cloud : Oracle Cloud Infrastructure (OCI), AWS, Terraform, CI/CD pipelines
Infrastructure As Code : Terraform, Ansible, Chef, Docker, Kubernetes, Jenkins, Linux, Git
Backend & Platform : REST APIs, distributed systems, control planes, serverless, event-driven architectures, Redis, SQL, infrastructure automation, reliability engineering, incident tooling
Monitoring : Grafana, MQL, JQL, Datadog
AI & Search Systems : RAG, vector search, BM25, Lucene, embeddings, agentic workflows

EXPERIENCE

Oracle India Development Center — 2020 – Present

Member of Technical Staff — India

- Reduced OCI API Gateway new-region and new-realm build time from **30 days to 8 hours** by redesigning the build workflow, replacing ticket-driven dependencies with reusable Terraform modules delivered through a CI/CD pipeline, and streamlining policy serialization and tenancy-aware provisioning.
- Led ARM adoption and mixed-shape platform validation across **34 production regions in 4 rollout phases**, enabling API Gateway services to run across x86 and ARM while addressing capacity bottlenecks through performance testing, shard sizing, and staged production rollout.
- Cut state-tier capacity footprint by **68.4% in OC1**, eliminating **357 cores** and reducing large-region usage from **63 OCPUs to 18 OCPUs per region**, by building an 8-step automated live-migration workflow for Redis-based state services with strict sequencing, health checks, and production guardrails.
- Re-architected the Orchestrator admin plane with a production-ready V2 API surface spanning **16 named resources**, modernized pagination and query behavior, decoupled API contracts from persistence through a DTO layer, and enabled parallel V1/V2 migration without underlying data-model changes.
- Cut access- and setup-related on-call pages from **10+ per week per UK–India shift to 0** by exposing secure public management APIs, removing tunnel dependencies, and enforcing tenant validation, RBAC, IP-based restrictions, and mTLS ingress.
- Built an internal operations copilot for runbooks, documentation, PDFs, HTML, tables, and multi-team knowledge bases, keeping retrieval latency **under 9 seconds** and reducing indexing time from **hours to minutes** through hybrid BM25/vector search, citation-aware responses, batch embeddings, and incremental indexing.
- Saved **203.5 engineering hours** within API Gateway and improved incident resolution time by **20%** by automating Jira alarm-context enrichment across **2,442 tickets**; onboarded **6 teams** to the workflow.
- Saved approximately **240 developer hours per month** across **15 teams** by launching an AI-assisted weekly operations reporting pipeline that clusters incidents, summarizes recurring themes, generates HTML reports, and publishes standardized outputs automatically.
- Modernized the API Gateway performance-testing stack by migrating the entire suite from legacy Canary to Test Service, upgrading to **Java 21** and a new testing framework, and building the supporting infrastructure; retired legacy instances, resolved recurring security-review findings, and avoided more than **300 security tickets** across **34 production regions**.

UnitedHealth Group 2017 – 2020
IO Engineering Analyst India

- Automated end-to-end Pega installation with Python and shell scripting, cutting per-server setup time from **11 hours to 30 minutes** and removing approximately **95%** of manual effort; recognized with the **Aquamarine Bravo Award for Innovation**.
- Built customized containerized Pega images on JBoss and Tomcat with MySQL for OpenShift deployment, advancing the platform's containerization strategy and enabling repeatable environment provisioning.
- Designed and deployed a three-tier high-availability web application using Apache HTTPD, Tomcat, and Prevayler, with auto-scaling and configuration management delivered through Docker, Docker Swarm, and Ansible.
- Saved **more than 30 hours of manual work per month** by automating certificate rotation, renewal notifications, and roll-out workflows with Ansible and Python.
- Cut go-live environment validation time by **2.5 hours per environment** by automating health checks across JBoss, WebSphere, and IHS with shell scripts that generated spreadsheet-based readiness reports.

KEY ACHIEVEMENTS

- Scaled Ops Copilot from a passion project into a senior-leadership-recognized operations platform adopted by **15 teams**, driving **hundreds of hours of monthly time savings** through AI-assisted troubleshooting, ticket enrichment, and reporting automation.
- Turned an API Gateway Catalog proof of concept into the foundation for a broader API Developer Portal initiative, strengthening internal developer enablement and platform self-service.
- Recognized with UnitedHealth Group's **Aquamarine Bravo Award for Innovation** for building a repeatable Pega installation workflow that replaced a lengthy manual process with fast, standardized server provisioning.

EDUCATION

Dayananda Sagar College of Engineering 2013 – 2017
B.E. in Information Science Bengaluru, India